

Traversing Arrays With Loops

Generated Aug 18, 2026 7:56 AM

What We'll Learn

By the end of this lesson, students can:

- Traverse an array using a for loop
- Access each array element using an index
- Display all values stored in an array
- Use an enhanced for loop to read array values
- Calculate the total of numerical array elements

Before We Start

Suppose an array contains the scores of five students.

```
int[] scores = {85, 90, 78, 92, 88};
```

How can we display every score without writing a separate `System.out.println()` statement for each element?

Let's Understand

Traversing an array means visiting each element of the array one at a time.

A loop allows us to process all array elements without repeating the same statement manually.

Using a for Loop

A regular for loop uses an index to access each array element.

```
int[] scores = {85, 90, 78, 92, 88};

for (int index = 0; index < scores.length; index++) {
    System.out.println(scores[index]);
}
```

The loop works as follows:

- index begins at 0, which is the first array position.
- index < scores.length keeps the index within the array.
- index++ moves to the next position.
- scores[index] accesses the element at the current position.

The length property contains the number of elements in an array.

```
System.out.println(scores.length);
```

Output:

```
5
```

The last valid index is always one less than the array's length.

```
Array length: 5
Valid indexes: 0, 1, 2, 3, 4
```

Using an Enhanced for Loop

Java also provides an enhanced for loop, sometimes called a for-each loop.

```
for (int score : scores) {
    System.out.println(score);
}
```

During each repetition, the variable score receives the value of the next array element.

Use an enhanced for loop when you only need to read each value and do not need its index.

Let's Practice

Create a Java program that stores these product prices in an array:

```
25, 40, 15, 60, 30
```

Your program must:

1. Declare and initialize the array.
2. Use a regular for loop to display each index and price.
3. Calculate the total of all prices.
4. Display the final total.

Expected output:

```
Index 0: 25  
Index 1: 40  
Index 2: 15  
Index 3: 60  
Index 4: 30  
Total: 170
```

After completing the program, replace the regular for loop with an enhanced for loop and display every price again.